

Team 2 – GreenStep Capstone

Department of Computer and Information Sciences

University of St. Thomas

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github.com/CISC-480-GreenStep/sp26_team2_greenstepchallenge_admin

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And to future student teams

Department of Computer and Information Sciences

University of St. Thomas

Dear Kristin and future student teams,

This letter accompanies the final deliverables for the GreenStep Challenge Admin Console, the spring 2026 CISC 480 capstone project at the University of St. Thomas. We are writing both to summarize what we built and to make it as easy as possible for whoever picks the work up next to keep going.

Project summary and impact. MPCA runs annual sustainability challenges that ask staff to log everyday actions across food, energy, transportation, water, and consumption categories. Before this project, that program lived in spreadsheets and email. The GreenStep Admin Console replaces that workflow with a single web application backed by a managed Postgres database. The application lets MPCA administrators design challenges, organize participants into flexible groups (departments, cohorts, building-level competitions), monitor engagement on a customizable drag-and-drop dashboard, and export participation reports for leadership. The console runs at <https://greenstep.auriga.fyi> and exposes three role tiers (SuperAdmin, Admin, GeneralUser) enforced at the database layer through Postgres Row-Level Security so authorization cannot be bypassed from the browser.

Constraints, limitations, and next steps. The capstone reached a working v0.9.0 build, but we want to be transparent about what is *not* in the box. The standalone Reports page and the new Participation Report dashboard widget still coexist; consolidating them under a single “Report and Analysis” surface is GitHub issue #40 and is partially complete. Two stub tables, **badges** and **tickets**, were planned but not finished; their UI shells exist on a closed pull request (#80) and are tracked in issue #90. The legacy **presets** / **preset_actions** tables remain on the live database even though no application code reads from them (issue #88). A few schema-versus-UI inconsistencies around challenge color fields are documented in issue #89. Photo moderation, push notifications, and multi-device dashboard layout sync are noted as future work in the Project Report. The mobile application built by Team 1 shares the same database but the two surfaces have not been exercised together at scale.

Implementation and continuity guidance. The repository is structured to be readable on first contact. The top-level `README.md` explains the project and how to run it; the `docs/` folder holds the architecture map, the consolidated product documentation (PRD, UX spec, technical design, source-code docs, QA activities), and this project report. Coding standards live in `CODING_GUIDELINES.md` and are enforced by ESLint and a 300-line-per-file cap. The database schema is consolidated into two migration files (`001_initial_schema.sql` and `002_seed_data.sql`); we squashed fourteen historical migrations into this baseline so the next team has one place to look, not fourteen.

We would suggest three concrete starting moves for the next team: (1) run `supabase db reset` locally against the consolidated baseline and confirm the schema matches production; (2) walk through issues #40, #88, #89, and #90 in that order to land the small hand-off items the team already mapped out; and (3) book an early meeting with Kristin to confirm whether the standalone Reports page should be retired before any new features are added.

Gratitude. Kristin, thank you for the time, candor, and flexibility you gave us across the semester. The willingness to walk us through how the agency actually runs these challenges, the patience as our design shifted from a FastAPI backend to Supabase, and the steady focus on the people the program is for made this a project we are proud to put our names on. The application exists because you were willing to treat four student developers as real collaborators, and we hope the next team gets the same opportunity.

Sincerely,

Josh Dunlap

Eli Goldberger

Khue Vo

Rudy Vergara

Team 2 – CISC 480 Senior Capstone, Section 01 (Spring 2026)
University of St. Thomas